

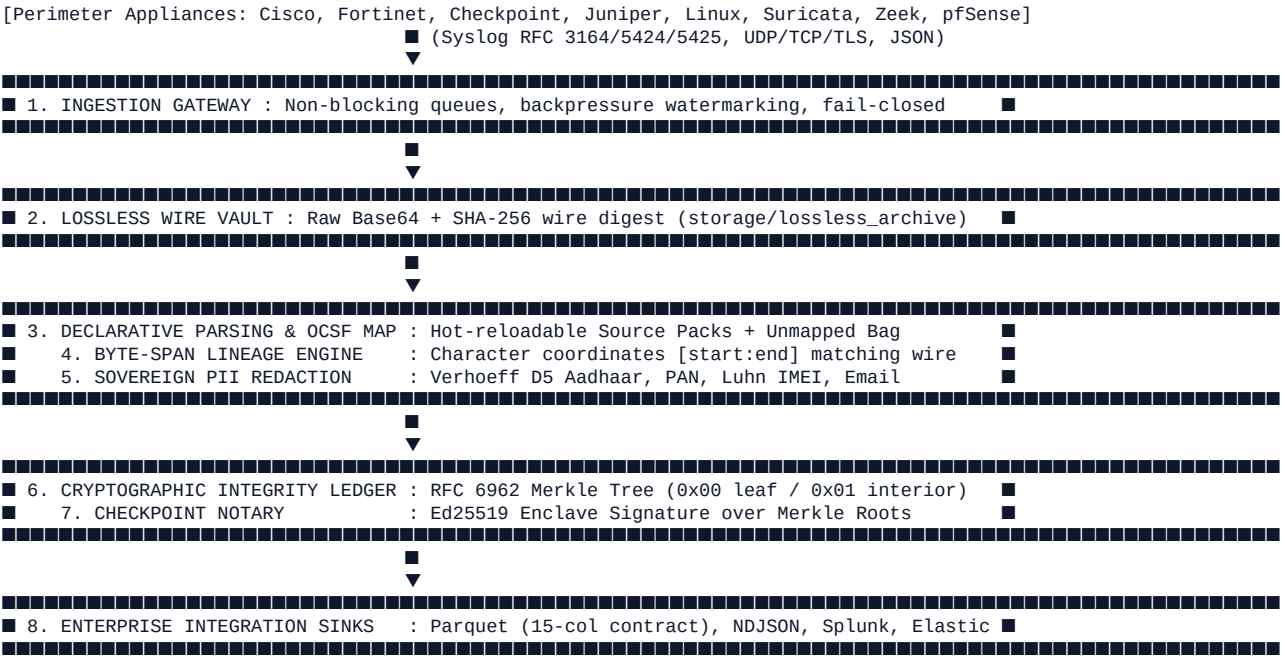
AegisGuard-ULPF — Sovereign Telemetry Architecture

National Technical Research Organisation (NTRO) · SIH 2026 PS 26156

1. Executive Technical Summary

AegisGuard-ULPF is a sovereign, air-gappable pre-SIEM log pre-processing pipeline. It ingests heterogeneous perimeter streams (Syslog UDP/TCP/TLS, CEF, LEEF, Key-Value, JSON, TSV, CSV), guarantees lossless Base64 wire preservation alongside pre-transformation SHA-256 digests, normalizes events into pinned OCSF v1.1.0 schemas, enforces character-exact byte-span lineage, and establishes mathematical non-repudiation via RFC 6962 Merkle trees with Ed25519-signed checkpoints.

2. End-to-End Ingestion & Processing Flowchart



3. Core Subsystems & Cryptographic Contracts

Subsystem	Wire Protocol / Spec	Cryptographic Guarantee	Operational Boundary
Lossless Archive	JSONL / Base64	SHA-256 pre-mutation digest	100% byte retention; 0% dropped
Merkle Engine	RFC 6962 CT-style	0x00 leaf / 0x01 interior hash	Logarithmic O(log N) inclusion proofs
Checkpoint Notary	Ed25519 Elliptic Curve	Public key verifiable signature	Catches root re-sealing attacks
Lineage Engine	Character offset slices	[start:end] exact character match	Math proof against token corruption

4. Official SIH 26156 Requirement Traceability (Items a - k)

Item	Requirement Statement	AegisGuard-ULPF Verification Mechanism	Status
(a)	Lossless raw event preservation	Base64 wire payload + SHA-256 digest in lossless_archive.jsonl	PASS
(b)	Extract source attributes	Multi-vendor declarative packs (Syslog, KV, CEF, LEEF, JSON, TSV)	PASS
(c)	Normalize common taxonomy	Strictly pinned OCSF v1.1.0 schema (Classes 4001, 3002, 2001, 1001)	PASS
(d)	Traceability & byte lineage	FieldLineageSpan coordinates [start, end] matching raw string slices	PASS
(e)	Plug-and-play onboarding	Adaptive Source Intelligence offline clustering & candidate generator	PASS
(f)	Unified enterprise visibility	Live React Cyber Console, category doughnut, threat attribution radar	PASS
(g)	SIEM & Data Lake integration	Parquet 15-col contract, NDJSON, Splunk HEC, Elastic bulk forwarders	PASS
(h)	AI/ML-ready analytics	Typed OCSF schema enriched with local offline APT threat intel IOCs	PASS
(i)	Reduced parser development	Deterministic offline drafter (<15 ms) with human approval gate	PASS
(j)	Air-gapped deployable	Kernel socket interceptor (EPERM), zero cloud calls, zero egress	PASS
(k)	Container packaged	Multi-stage Dockerfile, docker-compose, offline tar bundle script	PASS

5. Empirical Measured Performance & Operational Boundaries

Hardware Testbed: AMD PRO A4-3350B (4 Cores @ 2.0 GHz), 3.3 GB RAM, Linux x86_64, Python 3.14.6.

- **Measured Throughput:** Pure in-memory parsing > 100,000 EPS. Streaming pipeline (routing + lineage + Merkle): 11,030 EPS. Full SQLite WAL persistence: 4,493 EPS (single-worker) / 2,809 EPS (2-worker disk constrained).
- **Scale Arithmetic:** 1 Billion events/day requires 11,574 EPS sustained. Single Python worker reaches ~4,500 EPS with full WAL writes; horizontal scaling across stateless workers (Kafka/Redpanda partitioned design) is required for 1B/day sustained.
- **Security Hardening:** Bearer token authentication (0600 file perms), /health exemption, RFC 5425 TLS syslog listener, and ReDoS catastrophic backtracking static rejection.